NAME: STUDENT ID:

Question 1

Use the `gapminder` data frame in the `gapminder` library to recreate a version of Hans Rosling's famous data visualization shown in the *A Grammar of Graphics* slides (a single plot instead of a movie; in other words, for just a single year).

- You can see at a glance which years are available by running `count(gapminder, year)` in an R chunk.
- **Trim the data frame down to only a year's worth of observations.** You may use the following `dplyr` command. In this instance, we've picked the year 1987, but you can use any year in the data frame that you'd like!

```
gapminder_one_year <- filter(gapminder, year == 1987)
```

- In place of fertility rate use GDP per capita.

Constructing this plot requires several distinct steps.

1. Determine the correct geometry for the plot and make an initial `ggplot` with the two variables on the x and y axis.
2. Distinguish the continents by either shape or color. Which ever one you do not use, *set* its value to something other than the default.

- `ggplot` supports most color options (for instance, "red", "blue", "green").
- In `ggplot`, shapes are identified by a *number* between 1 and 25.

3. Alter the x and y axis labels so that they're more descriptive than just the variable given names in `gapminder`.
4. Add an annotation that draws attention to a particular feature of the data (of your choosing).
5. Title your plot with a claim based on your data.
6. Apply a theme of your choosing.

Use RStudio to write the code and see your visualization. Once you are happy with it, handwrite your code in the space below.

Question 2

Revisit the plot you made for Lab 2 (Class Survey), Question 8. This time, incorporate at least 2 of the elements from the Communicating with Graphics section of the Grammar of Graphics tutorial to polish your plot into one that tells a more focused story. No need to copy that code here, but in the space below, write a list of the elements that you applied to each plot.